

Three statistical questions are shown in the table. For each question, identify the type of data each question would produce, and a method that can be used to collect truthful data.

	Statistical Question	Type of Data	Method to Collect Truthful Data
1.	How many pets does a typical sixth grader have?	☐ categorical ☒ quantitative	Survey random students in 6 th grade around a specific school or district or city.
2.	What is the most popular style of music?	☒ categorical ☐ quantitative	Survey people of different ages randomly selected in a specific neighborhood or city or county.
3.	How long does it take to get to school in the morning?	☐ categorical ☒ quantitative	Survey students randomly selected that use different modes of transportation to arrive at school (bus, car, walk, bike, etc.)

Mila collected data from students in her class to answer the question: "How many hours per week do you spend on homework?" Her data are shown.

2	3.5	3	8	1.5	5	4.5	3	2.5	10
6.5	7.5	2	8	6	7.5	2.5	4.5	3	3.5

1. Determine the five-number summary for Mila's data.

Minimum: 1.5

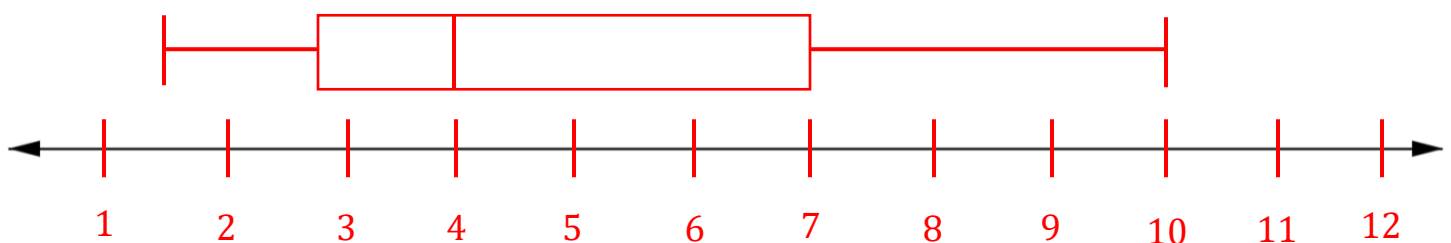
First Quartile: 2.75

Median: 4

Third Quartile: 7

Maximum: 10

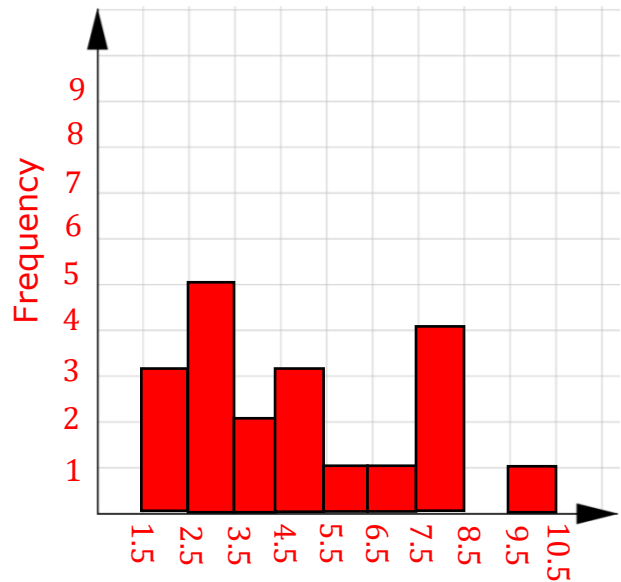
2. Create a box plot to display Mila's data.



3. Create a frequency chart of the data.

Hours	Frequency
1.5 – 2.5	3
2.5 – 3.5	5
3.5 – 4.5	2
4.5 – 5.5	3
5.5 – 6.5	1
6.5 – 7.5	1
7.5 – 8.5	4
8.5 – 9.5	0
9.5 – 10.5	1

7. Create a histogram of the data.



Range: $10 - 1.5 = 8.5$; $\frac{8.5}{8 \text{ intervals}} = 1.0625 \rightarrow$ made a 9th row b/c rounded interval width to 1

8. What is the mean of Mila's data?

$$(2 + 3.5 + 3 + 8 + 1.5 + 5 + 4.5 + 3 + 2.5 + 10 + 6.5 + 7.5 + 2 + 8 + 6 + 7.5 + 2.5 + 4.5 + 3 + 3.5) \div 20 = 94 \div 20 = 4.7$$

The average amount of time students spend on homework each week is 4.7 hours.

9. What is the median of Mila's data?

4 meaning half of the students studied for more than 4 hours and half studied for less than 4 minutes.

10. Summarize Mila's results using statistical terminology.

The graph is right skewed and has a gap between 8.5 and 9.5. Most of the students spent between 1.5 and 4.5 hours on homework.

The mean of 4.7 means the average time students spent on homework was 4.7 hours and correlates to the median of 4.

The median means half of the students spent less than 4 hours on homework, and the other half of the class spent more than 4 hours on homework.